Gwangwoong Park

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gwangwoongpark

Education

BS **Sogang University**, Business Administration and Computer Science and Engineering (Double Major) — Seoul, South Korea, Mar 2023 – Feb 2027

- Expected graduation: Feb 2027
- Presidential Award for Academic Excellence (Fall 2025, top 1% semester GPA)
- Dean's List (Fall 2024, Fall 2025; top 3% semester GPA)
- Beta Gamma Sigma (2025) and Alpha Sigma Nu (2025) honor societies

Experience

NLP & ISDS Lab, Sogang University, Undergraduate Research Intern (Advisor: Prof. Du-Seong Chang) — Seoul, South Korea, Jan 2026 – present, 11 months

- Research interests: multimodal understanding, representation learning, generative models, and vision-language-action models
- Lab seminars: an overview of diffusion models, and audio-aware robot policies (MultiGen, HEAR)

Publications

Automatic Multi-Track Sound Metadata Generation via Hierarchical Video Analysis Using Vision Language Models — June 2026

Gwangwoong Park, Youngwoo Cho, Suhwan Tcha, Jaewon Choi, Du-Seong Chang
www.dbpia.co.kr/journal/articleDetail?nodeId=NODE12930030 (Korea Computer Congress (KCC 2026), pp. 2466–2468)

Projects

SoundPilot — Sound Design Agent for Multi-Track Audio — Spring 2026

- Capstone Design project. Defined the problem and designed the agent pipeline: scene split, VLM object identification, SAM 3 tracking, tool-calling agents on a shared sound timeline, and routing to video-to-audio, text-to-audio, and text-to-speech models
- Measured run-to-run variance of VLM cut detection and replaced it with a deterministic detector for stable results
- Gold Prize (1st place) at the Sogang Capstone Design Competition; KCC 2026 accepted (first author); Main Award Finalist at the AI·SW-Centered University Excellent Project Competition

Assistive Board-Game Robot (in progress) — Sep 2026 – present

- Team lead. Funded by Sogang's Creative Convergence Research Program (KRW 1.2M)
- Designed the hardware and ordered parts: SO-101 leader and follower arms, top and wrist cameras, and gripper FSR tactile sensors read by an ESP32
- Plan: VLA fine-tuning on teleoperated demonstrations, staged-reward RL in Isaac Lab, and sim-to-real transfer; voice commands for users with limited hand mobility

Airport Delay Network — Nov 2025

- CAU × Sogang University Data Hackathon. Led insight discovery: modeled delay propagation across airports with PageRank and visualized it; Excellence Award

Movie Recommendation with Item Embeddings 2025

- Basic Recommender Systems course (A+). Learned MovieLens movie embeddings from co-watched pairs with genre information, and recommended unseen movies by cosine similarity to a user taste vector

Squat Form Assessment 2025

- Parrot (Sogang University Data Science Society). Compared XGBoost, an LSTM on joint-angle sequences, and a video ViT; built the MediaPipe test pipeline for real user videos, handled occluded joints, and made a web demo with per-rep scores and feedback

Diffusion-Based Dataset Distillation Fall 2025

- AI Data Analysis, joint undergraduate–graduate course (A+). Built a live MNIST demo with the original method's code and released distilled images (ten images train a LeNet to 88.7%), and compared trajectory matching with diffusion-based distillation (D4M)

Honors

Nov 2026: AI·SW-Centered University Excellent Project Competition — Main Award Finalist as Sogang's representative (18 of 60 teams)

Spring 2026: Sogang Capstone Design Competition — Gold Prize (1st place)

Fall 2025: Presidential Award for Academic Excellence, Sogang University

Nov 2025: CAU × Sogang University Data Hackathon — Excellence Award

Teaching and Mentoring

Fall 2025: Teaching Assistant, Introduction to AI Programming, Sogang University

Fall 2025: Mentor, Parrot (Sogang University Data Science Society)

2024: Online Mentor, KT LAN-Night School

Oct 2023 – Feb 2025: Education volunteer at a local children's center, KOSAF scholarship program (439 hours)